

AIMS 5740

Generative Artificial Intelligence

(2026 Term 2)



Computer Science & Engineering
The Chinese University of Hong Kong

| Grading

2 x 15% Two programming assignments

30% Midterm Quiz (scheduled on
March 18, 2026 7:00 PM)

40% Final Project (report+presentation)

| Announcement

- The exam is scheduled for 2 hours.
- You can **only** bring basic calculator and the course slides.
- No cheating. No talking to each other.

Basic

- Text:
 - Contextual embedding
 - **Transformer**
 - Bert, **GPT**
- Image:
 - CNN
 - Diffusion process
 - DDPM, Stable Diffusion

| Data & Compute (1)

- Data:
 - Data resource
 - Data processing: deduplication, model-based selection
 - Data augmentation: back translation, LLM-based augmentation
 - Evaluation benchmark, Leaderboard
- Compute:
 - DNN Platform: Pytorch, Tensorflow
 - **Data Parallelism, Pipeline Parallelism, Tensor Parallelism**
 - Mixed Precision Training, Gradient Checkpointing/Accumulation
 - Metrics for LLM Inference
 - Flash attention, KV caching
 - Speculative decoding

| Data & Compute (2)

- Data & Compute:
 - **Scaling laws**
 - Improved scaling laws
 - Prompt engineering, In-context learning

| Modeling Architecture

- Mixture of Experts:
 - **Basic architecture**
 - **Loss function**, training, **Expert Parallelism**
 - Sparse upcycling, Sparse splitting
- Linear Recurrent Models:
 - State space Models, Mamba
 - Linear attention
- **Multimodal Learning:**
 - MLLMs: Flamingo, LLaVA, CLIP
 - Unified models: Janus-Pro, Qwen-image, Bagel